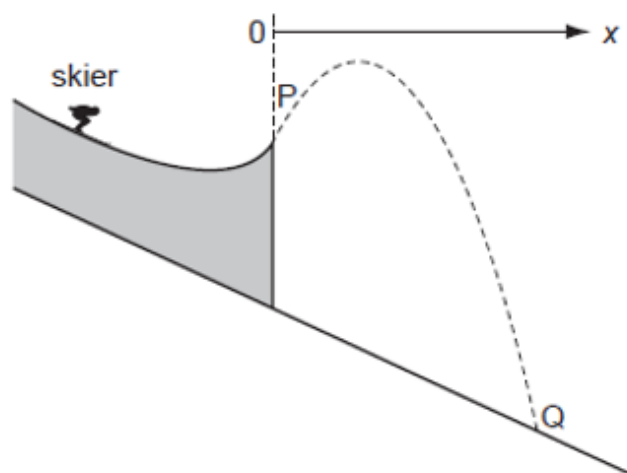
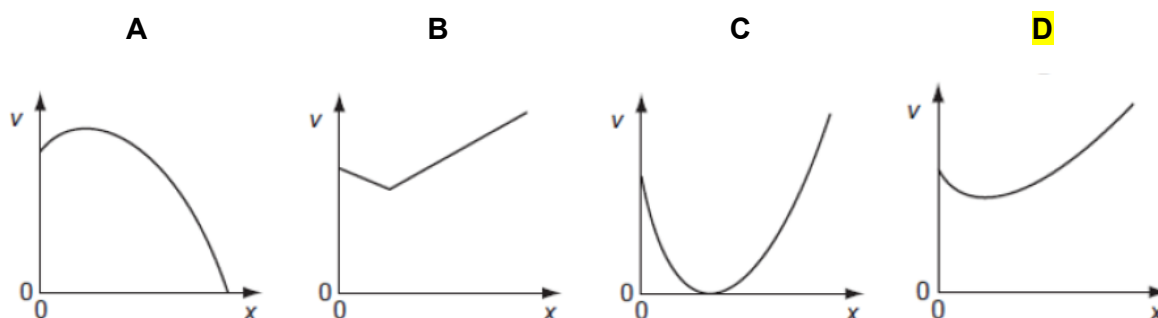


4. The dotted line shows the path of a competitor in a ski-jumping competition.



Ignoring air resistance, which graph best represents the variation of his speed v with the horizontal distance x covered from the start of his jump at P before landing at Q?



Answer: D

- speed v decreases from P to highest point and increases from highest point
 - v cannot be zero due to horizontal velocity is constant
- At this point, you may narrow your options to B and D*

Quantitative

Let u_x be horizontal speed and u_y be vertical speed at P. and v be speed between P and Q. Let h be the vertical displacement from P.

Using energy conservation,

$$\Delta E_k + \Delta E_p = 0 \Rightarrow \frac{1}{2}mv^2 - \frac{1}{2}mu^2 + mgh = 0$$

$$v^2 = u^2 - 2gh \quad \dots\dots (1)$$

vertically: $h = u_y t - \frac{1}{2}gt^2 \quad \dots\dots (2)$

horizontally: $x = u_x t \quad \square \quad t = \frac{x}{u_x} \quad \dots\dots (3)$

sub (3) into (2): $h = \frac{u_y}{u_x} x - \frac{g}{2u_x^2} x^2 \quad \dots\dots (4)$

sub (4) into (1): $v^2 = u^2 - \frac{2gu_y}{u_x} x + \frac{g^2}{u_x^2} x^2$

You can use graphing calculators to plot $y^2 = 1 - x - x^2$

